FOCUS Laying foundation for IIT-JEE
-An IIT(ISM) Alumni & Student's Undertaking

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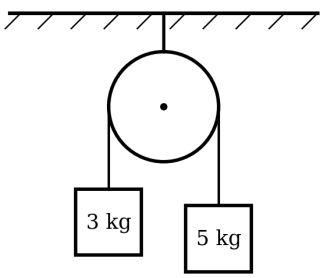
CLASS 9 - JEE MAIN PATTERN TEST

Time Duration: 3 hours **Maximum Marks:** 300
Total Questions: 75 (60 MCQs + 15 INTEGER TYPE)
Marking Scheme: +4 for each correct answer, -1 for each wrong answer

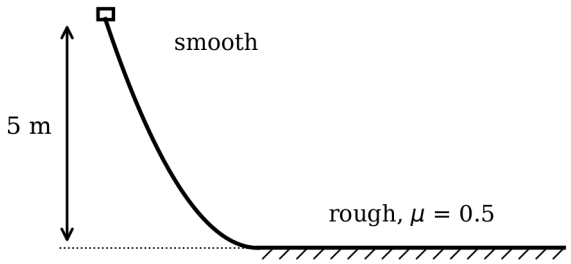
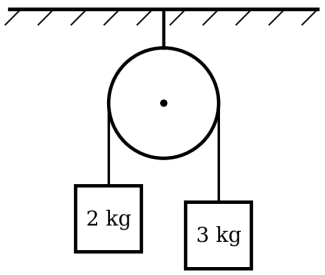
Take g = 10 m s^{-2}. Atomic masses: H = 1, C = 12, N = 14, O = 16, Na = 23, Al = 27, S = 32, Cl = 35.5, Ca = 40, Zn = 65. Molar volume at STP = 22.4 L; N_A = Avogadro number.

SECTION-A (MCQ)

PHYSICS

1.	The position of a particle moving along a straight line is $x = t^3 - 6t^2 + 9t$ (x in metre, t in second). The total distance travelled by the particle in the first 4 s is (a) 4 m (b) 12 m (c) 8 m (d) 16 m
2.	A ball is thrown vertically upward with a speed of 20 m s^{-1} from the top of a 25 m high tower. It reaches the ground after a time of (g = 10 m s^{-2}) (a) 4 s (b) 3 s (c) 6 s (d) 5 s
3.	Two masses of 3 kg and 5 kg are connected by a light string passing over a smooth light pulley as shown. The tension in the string is (g = 10 m s^{-2})  (a) 37.5 N (b) 30 N (c) 50 N (d) 40 N

4.	Two blocks of masses 2 kg and 3 kg are placed in contact on a smooth horizontal surface. A horizontal force of 10 N is applied on the 2 kg block, pushing it against the 3 kg block. The contact force between the blocks is (a) 4 N (b) 10 N (c) 6 N (d) 5 N
5.	A force $\vec{F} = (2\hat{i} + 3\hat{j})$ N displaces a particle from the origin to the point (4, −1) m. The work done by the force is (a) 11 J (b) 8 J (c) 3 J (d) 5 J
6.	A block of mass 2 kg slides 10 m down a smooth incline of angle 30°. The work done by gravity on the block is (g = 10 m s^{-2}) (a) 200 J (b) 100 J (c) 50 J (d) 173 J
7.	A particle moves along the x-axis under a force F = ($3x^2 - 2x + 1$) N. The work done by the force from x = 0 to x = 2 m is (a) 10 J (b) 4 J (c) 6 J (d) 8 J
8.	A bullet of mass 20 g moving at 300 m s^{-1} penetrates 5 cm into a fixed wooden block and stops. The average resistive force offered by the block is (a) 18000 N (b) 9000 N (c) 36000 N (d) 1800 N
9.	A car of mass 1000 kg has an engine of power 40 kW. If the resistive force on the car is a constant 1000 N, the maximum speed of the car is (a) 20 m s^{-1} (b) 40 m s^{-1} (c) 60 m s^{-1} (d) 30 m s^{-1}
10.	Two bodies of masses m and 4m have equal kinetic energies. The ratio of their linear momenta is (a) 1 : 4 (b) 4 : 1 (c) 2 : 1 (d) 1 : 2
11.	A ball of mass 0.5 kg is placed against a spring of force constant 200 N m^{-1} compressed by 0.2 m on a smooth horizontal floor. When released, the speed of the ball on leaving the spring is (a) 4 m s^{-1} (b) 2 m s^{-1} (c) 8 m s^{-1} (d) $2\sqrt{2}$ m s^{-1}

12.	A block is released from rest from a height of 5 m on a smooth curved track which ends on a rough horizontal surface with $\mu = 0.5$, as shown. The distance travelled by the block on the rough surface before it stops is  (a) 5 m (b) 20 m (c) 10 m (d) 2.5 m
13.	The potential energy of a particle moving along the x-axis is $U = 2x^3 - 3x^2$ (in joule, x in metre). The force acting on the particle at x = 2 m is (a) 12 N (b) −24 N (c) 0 N (d) −12 N
14.	A body of mass 2 kg, initially at rest, is acted upon by a constant force of 4 N. The instantaneous power delivered by the force at t = 5 s is (a) 20 W (b) 40 W (c) 10 W (d) 80 W
15.	A ball is projected vertically upward with a speed of 20 m s^{-1}. At what height above the point of projection are its kinetic energy and potential energy equal? ($g = 10$ m s^{-2}) (a) 20 m (b) 5 m (c) 10 m (d) 15 m
16.	In the arrangement shown, the 2 kg and 3 kg blocks are released from rest. The speed of the blocks after the heavier block has descended 1 m is ($g = 10$ m s^{-2})  (a) 2 m s^{-1} (b) 4 m s^{-1} (c) 1 m s^{-1} (d) $\sqrt{2}$ m s^{-1}
17.	A man pushes a 20 kg block up a rough incline of angle 37° ($\mu = 0.5$) through 5 m at constant speed, with the force parallel to the incline. The work done by the man is ($\sin 37° = 0.6$, $\cos 37° = 0.8$, $g = 10$ m s^{-2}) (a) 600 J (b) 1000 J (c) 800 J (d) 400 J

18.	A motor of input power 5 kW and efficiency 80% is used to lift water from a well 20 m deep. The mass of water lifted per second is (g = 10 m s^{-2}) (a) 25 kg s^{-1} (b) 16 kg s^{-1} (c) 40 kg s^{-1} (d) 20 kg s^{-1}
19.	A 2 kg block moving at 6 m s^{-1} collides with a stationary 4 kg block and the two stick together. The loss of kinetic energy in the collision is (a) 24 J (b) 12 J (c) 36 J (d) 18 J
20.	The work done in stretching a spring from its natural length by 2 cm is W. The work required to stretch it further from 2 cm to 6 cm is (a) 4W (b) 9W (c) 8W (d) 3W

CHEMISTRY

21.	Which of the following compounds contains both ionic and covalent bonds? (a) HCl (b) CH_4 (c) $MgCl_2$ (d) NaOH
22.	The total number of shared electron pairs in one molecule of CO_2 is (a) 2 (b) 4 (c) 3 (d) 6
23.	Which of the following molecules does not obey the octet rule at its central atom? (a) CH_4 (b) NH_3 (c) BF_3 (d) H_2O
24.	An element X (atomic number 12) combines with an element Y (atomic number 17). The formula and nature of the compound formed is (a) XY_2, ionic(b) XY, covalent (c) X_2Y, ionic(d) XY_2, covalent
25.	10 g of $CaCO_3$ on heating gives 5.6 g of CaO and a gas. According to the law of conservation of mass, the mass of the gas evolved is (a) 5.6 g (b) 4.4 g (c) 10 g (d) 4.0 g
26.	A sample of a compound contains 4 g of A and 1 g of B. According to the law of constant proportions, the mass of B present in another sample of the same compound containing 12 g of A is (a) 4 g (b) 2 g (c) 6 g (d) 3 g
27.	Nitrogen forms N_2O and NO_2. The ratio of the masses of oxygen that combine with a fixed mass of nitrogen in N_2O and NO_2 is (N = 14, O = 16) (a) 1 : 4 (b) 1 : 2 (c) 1 : 1 (d) 2 : 1
28.	The number of electrons present in 2.8 g of N_2 is (N = 14; N_A = Avogadro number) (a) 0.7 N_A (b) 2.8 N_A (c) 1.4 N_A (d) 0.14 N_A

29.	The mass of one molecule of water is approximately ($N_A = 6.022 \times 10^{23}$) (a) 1.8×10^{-22} g(b) 6.0×10^{-26} g (c) 3.0×10^{-26} g(d) 3.0×10^{-23} g
30.	A gaseous mixture contains 4 g of He and 16 g of CH_4. The average molar mass of the mixture is (He = 4, C = 12, H = 1) (a) 8 g mol^{-1} (b) 10 g mol^{-1} (c) 20 g mol^{-1} (d) 12 g mol^{-1}
31.	The volume occupied at STP by 6.022×10^{22} molecules of a gas is (a) 22.4 L (b) 0.224 L (c) 2.24 L (d) 224 L
32.	4 mol of Al reacts with 3 mol of Cl_2 according to $2Al + 3Cl_2 \rightarrow 2AlCl_3$. The number of moles of $AlCl_3$ formed is (a) 2 (b) 4 (c) 3 (d) 1.5
33.	12 g of carbon is burnt in 64 g of oxygen according to $C + O_2 \rightarrow CO_2$. The mass of CO_2 formed is (C = 12, O = 16) (a) 88 g (b) 44 g (c) 22 g (d) 76 g
34.	$Zn + 2HCl \rightarrow ZnCl_2 + H_2$. The volume of H_2 evolved at STP when 13 g of Zn reacts with excess HCl is (Zn = 65) (a) 2.24 L (b) 8.96 L (c) 22.4 L (d) 4.48 L
35.	25 g of impure $CaCO_3$ on strong heating gives 8.8 g of CO_2. The percentage purity of the sample is (Ca = 40, C = 12, O = 16) (a) 80% (b) 88% (c) 70% (d) 90%
36.	The number of moles of O_2 required for the complete combustion of 1 mol of butane (C_4H_{10}) is (a) 13 (b) 4 (c) 6.5 (d) 9
37.	The molality of a solution containing 18 g of glucose ($C_6H_{12}O_6$) dissolved in 200 g of water is (C = 12, H = 1, O = 16) (a) 0.1 m (b) 1.0 m (c) 0.9 m (d) 0.5 m
38.	A 500 g sample of water contains 0.0025 g of dissolved fluoride. The concentration of fluoride in ppm (by mass) is (a) 0.5 ppm (b) 5 ppm (c) 50 ppm (d) 0.05 ppm
39.	The molarity of 36.5% (w/w) HCl solution of density 1.2 g mL^{-1} is (H = 1, Cl = 35.5) (a) 10 M (b) 36.5 M (c) 12 M (d) 1.2 M
40.	The mole fraction of the solute in a 1.00 molal aqueous solution is (a) 0.0177 (b) 0.0556 (c) 0.177 (d) 0.0018

41.	The value of $\log_2 32 + \log_3 81 - \log_{10} 1000$ is (a) 5 (b) 6 (c) 7 (d) 12
42.	If $\log_{10} 2 = 0.3010$, the number of digits in 2^{20} is (a) 6 (b) 8 (c) 5 (d) 7
43.	If $x = \log_a (bc)$, $y = \log_b (ca)$ and $z = \log_c (ab)$, then $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}$ = (a) 1 (b) 0 (c) abc (d) 2
44.	The solution of $\log_2 (x-1) + \log_2 (x+1) = 3$ is (a) $x = -3$ (b) $x = \pm 3$ (c) $x = 3$ (d) $x = 9$
45.	If $\log_4 x + \log_2 x = 6$, then x = (a) 8 (b) 4 (c) 64 (d) 16
46.	The value of $\log_{25} 125$ is (a) $\frac{5}{2}$ (b) $\frac{3}{2}$ (c) $\frac{2}{3}$ (d) 5
47.	The number of positive integers x satisfying $\log_3 (x-1) \le 2$ is (a) 8 (b) 10 (c) 9 (d) 11
48.	If $\log_{10} 3 = a$ and $\log_{10} 5 = b$, then $\log_{30} 8$ = (a) $\frac{3(1-b)}{1+a}$ (b) $\frac{3(1+b)}{1+a}$ (c) $\frac{3(1-b)}{1-a}$ (d) $\frac{3b}{1+a}$
49.	If $\log_x (0.1) = -\frac{1}{3}$, then x = (a) 100 (b) 1000 (c) 10 (d) 0.001
50.	The characteristic of $\log_{10} 0.0035$ is (a) −2 (b) 3 (c) 2 (d) −3
51.	The value of $(\log_2 3)(\log_3 4)(\log_4 5)(\log_5 8)$ is (a) 3 (b) 2 (c) 4 (d) 1

52.	Two vectors of magnitudes 5 and 12 act at right angles to each other. The magnitude of their resultant is (a) 7 (b) 17 (c) 13 (d) 10
53.	Two forces of 7 N and 3 N act at a point. Which of the following cannot be the magnitude of their resultant? (a) 4 N (b) 7 N (c) 10 N (d) 12 N
54.	If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$, then $\lvert\vec{a} + \vec{b}\rvert$ = (a) $\sqrt{6}$ (b) $\sqrt{14}$ (c) 14 (d) 6
55.	Three vectors of magnitudes 3, 4 and 5 units add up to zero. The angle between the vectors of magnitudes 3 and 4 is (a) 60° (b) 120° (c) 90° (d) 0°
56.	Two vectors of equal magnitude A have a resultant of magnitude A. The angle between them is (a) 120° (b) 60° (c) 90° (d) 150°
57.	Two perpendicular vectors of magnitudes 8 and 6 are added. The angle made by the resultant with the vector of magnitude 8 is (a) $\tan^{-1}\frac{4}{3}$ (b) $\tan^{-1}\frac{3}{4}$ (c) 45° (d) 30°
58.	If $\vec{a} = 5\hat{i} + 2\hat{j}$ and $\vec{b} = -2\hat{i} + 2\hat{j}$, the unit vector along $\vec{a} + \vec{b}$ is (a) $\frac{4\hat{i} + 3\hat{j}}{5}$ (b) $\frac{3\hat{i} + 4\hat{j}}{7}$ (c) $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$ (d) $\frac{3\hat{i} + 4\hat{j}}{5}$
59.	For two non-zero vectors, $\lvert\vec{A} + \vec{B}\rvert = \lvert\vec{A} - \vec{B}\rvert$. The angle between $\vec{A}$ and $\vec{B}$ is (a) 90° (b) 0° (c) 180° (d) 45°
60.	A force of 20 N acts at 60° to the horizontal. Its horizontal and vertical components are respectively (a) $10\sqrt{3}$ N, 10 N (b) 20 N, 0 N (c) 10 N, $10\sqrt{3}$ N (d) 10 N, 10 N

SECTION-B (INTEGER TYPE)

The answer to each question is a non-negative integer.

PHYSICS

61.	A particle starts from rest with acceleration a = 2t m s^{-2}. The distance travelled by it in the first 3 s is _____ m.
62.	A 2 kg block, initially at rest, is pulled up a smooth incline of angle 30° through 5 m by a force of 20 N acting parallel to the incline. The kinetic energy gained by the block is _____ J. (g = 10 m s^{-2})
63.	A lift of mass 900 kg moves upward at a constant speed of 2 m s^{-1} against a frictional force of 1000 N. The power delivered by the motor is _____ kW. (g = 10 m s^{-2})
64.	A block of mass 2 kg moving at 5 m s^{-1} on a smooth horizontal floor strikes a spring of force constant 200 N m^{-1} and compresses it. The maximum compression of the spring is _____ cm.
65.	A force F = (6x − 4) N acts on a particle along the x-axis. The work done by the force in moving the particle from x = 0 to x = 4 m is _____ J.

CHEMISTRY

66.	The number of moles of atoms present in 98 g of H_2SO_4 is _____. (H = 1, S = 32, O = 16)
67.	The mass of oxygen (in g) required for the complete combustion of 8 g of methane (CH_4) is _____. (C = 12, H = 1, O = 16)
68.	3 mol of N_2 reacts with 6 mol of H_2 according to $N_2 + 3H_2 \rightarrow 2NH_3$. The number of moles of NH_3 formed is _____.
69.	The volume (in mL) of 0.5 M H_2SO_4 required to completely neutralise 100 mL of 1 M NaOH is _____.
70.	The mass of water (in g) in which 36 g of glucose (molar mass 180 g mol^{-1}) must be dissolved to prepare a 0.5 molal solution is _____.

MATHEMATICS

71.	If $\log_2 x + \log_4 x + \log_8 x = 11$, then x = _____.
72.	The value of $100^{\log_{10} 3 + \log_{10} 2}$ is _____.
73.	If $\log_{10} 2 = 0.3010$, the number of digits in 5^{30} is _____.

74.	If $\vec{a} = 4\hat{i} + \hat{j}$, $\vec{b} = -3\hat{i} + 5\hat{j}$ and $\vec{c} = 2\hat{i} - 2\hat{j}$, then $\lvert\vec{a} + \vec{b} + \vec{c}\rvert$ = _____.
75.	Two vectors of magnitudes 3 and 5 have a resultant of magnitude 7. The angle between them is _____ degrees.